7 CHEMICAL COORDINATION

The nervous system (Chapter 6) is a coordination system forming a link between stimulus and response. The body has a second coordination system, which does not involve nerves. This is the endocrine system. It consists of organs called endocrine glands, which make chemical messenger substances called hormones. Hormones are carried in the bloodstream.

LEARNING OBJECTIVES

- Describe how responses can be controlled by hormonal communication
- Understand the differences between nervous and hormonal control
- Understand the sources, roles and effects of the following hormones:
  - adrenaline
  - insulin
  - testosterone*
  - progesterone*
  - oestrogen*

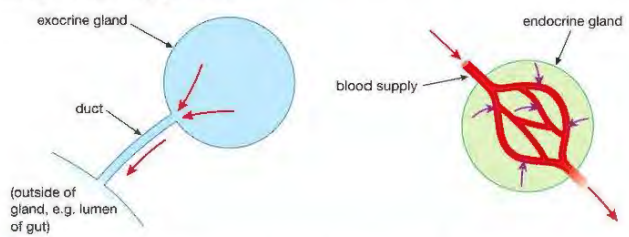
BIOLOGY ONLY

  - antidiuretic hormone (ADH)*
  - follicle stimulating hormone (FSH)*
  - luteinising hormone (LH)*

*These hormones will be dealt with in more detail in later chapters.

GLANDS AND HORMONES

A gland is an organ that releases or *secretes* a substance. This means that cells in the gland make a chemical which leaves the cells through the cell membrane. The chemical then travels somewhere else in the body, where it carries out its function. There are two types of glands – exocrine and endocrine glands. **Exocrine glands** secrete their products through a tube called a duct. For example, salivary glands in your mouth secrete saliva down salivary ducts, and tear glands secrete tears through ducts that lead to the surface of the eye. **Endocrine glands** have no duct, and so are called ductless glands. Instead, their products, the **hormones**, are secreted into the blood vessels that pass through the gland (Figure 7.1).



▲ Figure 7.1 Exocrine glands secrete their products though a duct, while endocrine glands secrete hormones into the blood.

This chapter looks at some of the main endocrine glands and the functions of the hormones they produce. Because hormones are carried in the blood, they can travel to all areas of the body. They usually only affect certain tissues or organs, called 'target organs', which can be a long distance from the gland that made the hormone. Hormones only affect particular tissues or organs if the cells of that tissue or organ have special chemical receptors for the particular hormone. For example, the hormone **insulin** affects the cells of the liver, which have insulin receptors.

KEY POINT

The receptors for some hormones are located in the cell membrane of the target cell. Other hormones have receptors in the cytoplasm, and some in the nucleus. Without specific receptors, a cell will not respond to a hormone at all.

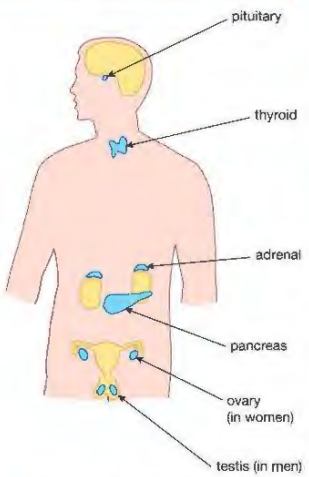
THE DIFFERENCES BETWEEN NERVOUS AND ENDOCRINE CONTROL

Although the nervous and endocrine systems both act to coordinate body functions, there are differences in the way that they do this. These are summarised in Table 7.1.

Table 7.1 The nervous and endocrine systems compared.

Nervous system	Endocrine system
works by nerve impulses transmitted through nerve cells (although chemicals are used at synapses)	works by hormones transmitted through the bloodstream
nerve impulses travel fast and usually have an 'instant' effect	hormones travel more slowly and generally take longer to act
response is usually short-lived	response is usually longer-lasting
impulses act on individual cells such as muscle fibres, so have a very localised effect	hormones can have widespread effects on different organs (although they only act on particular tissues or organs if the cells have the correct receptors)

THE ENDOCRINE GLANDS



▲ Figure 7.2 The main endocrine glands of the body.

KEY POINT

The pituitary is a link between the nervous and endocrine coordination systems.

The positions of the main endocrine glands are shown in Figure 7.2. A summary of some of the hormones that they make and their functions is given in Table 7.2.

The **pituitary gland** (often just called 'the pituitary') is found at the base of the brain. It produces a number of hormones, including **antidiuretic hormone (ADH)**, which acts on the kidneys, controlling the amount of water in the blood (see Chapter 8). The pituitary also releases hormones that regulate reproduction (see Chapter 9).

Just above the pituitary is a part of the brain called the **hypothalamus**. The pituitary contains neurones linking it to the hypothalamus, and some of its hormones are produced under the control of the brain.

Table 7.2: Some of the main endocrine glands, the hormones they produce and their functions.

Gland	Hormone	Some functions of the hormones
pituitary	follicle stimulating hormone (FSH) luteinising hormone (LH) antidiuretic hormone (ADH)	stimulates egg development and oestrogen secretion in females and sperm production in males stimulates egg release (ovulation) in females and testosterone production in males controls the water content of the blood
thyroid	thyroxine	controls the body's metabolic rate (how fast chemical reactions take place in cells)
pancreas	insulin glucagon	lowers blood glucose raises blood glucose
adrenals	adrenaline	prepares the body for physical activity
testes	testosterone	controls the development of male secondary sexual characteristics
ovaries	oestrogen progesterone	controls the development of female secondary sexual characteristics regulates the menstrual cycle

The **pancreas** is both an endocrine *and* an exocrine gland. It secretes two hormones involved in the regulation of blood glucose, and is also a gland of the digestive system, secreting enzymes through the pancreatic duct into the small intestine (see Chapter 4). The sex organs of males (**testes**) and females (**ovaries**) are also endocrine organs. In addition to their role in producing sex cells, the testes and ovaries make hormones that are involved in controlling reproduction. This topic is covered more fully in Chapter 9. We will now look at the functions of two hormones in more detail.

ADRENALINE – THE 'FIGHT OR FLIGHT' HORMONE

DID YOU KNOW?
'Adrenal' means 'next to the kidneys', which describes where the adrenal glands are located – on top of these organs (see Figure 7.2).

When you are frightened, excited or angry, your **adrenal glands** secrete the hormone adrenaline.

Adrenaline acts at a number of target organs and tissues, preparing the body for action. In animals other than humans this action usually means dealing with an attack by an enemy, where the animal can stay and fight or run away – hence 'fight or flight'. This is not often a problem with humans, but there are plenty of other times when adrenaline is released (Figure 7.3).



▲ Figure 7.3 Many human activities cause adrenaline to be produced, not just a 'fight or flight' situation!



▲ Figure 7.4 Adrenaline affects the body of an animal in many ways.

If an animal's body is going to be prepared for action, the muscles need a good supply of oxygen and glucose for respiration. Adrenaline produces several changes in the body that make this happen (Figure 7.4) as well as other changes to prepare for fight or flight.

- The breathing rate increases and breaths become deeper, taking more oxygen into the body.
- The heart beats faster, sending more blood to the muscles, so that they receive more glucose and oxygen for respiration.
- Blood is diverted away from the intestine and into the muscles.
- In the liver, stored carbohydrate (glycogen) is changed into glucose and released into the blood. The muscle cells absorb more glucose and use it for respiration.
- The pupils dilate, increasing visual sensitivity to movement.
- Body hair stands upright, making the animal look larger to an enemy.
- Mental awareness is increased, so reactions are faster.

In humans, adrenaline is not just released in a 'fight or flight' situation, but in many other stressful activities too, such as preparing for a race, going for a job interview or taking an exam.

INSULIN – CONTROL OF BLOOD GLUCOSE

You saw earlier that adrenaline can raise blood glucose from stores in the liver. The liver cells contain carbohydrate in the form of **glycogen**. Glycogen is made from long chains of glucose sub-units joined together, forming a large insoluble molecule (see Chapter 4). Being insoluble makes glycogen a good storage product. When the body is short of glucose, the glycogen can be broken down into glucose, which then passes into the bloodstream.

Adrenaline raises blood glucose concentration in an emergency, but other hormones act all the time to control the level, keeping it fairly constant at a little less than 1 g of glucose in every dm^3 (cubic decimetre) of blood. The main hormone controlling glucose is insulin. Insulin is made by special cells in the pancreas. It stimulates the liver cells to take up glucose and convert it into glycogen, lowering the level of glucose in the blood.

DID YOU KNOW?
The pancreas also makes a hormone called **glucagon**, which has the opposite effect to insulin – it raises blood glucose.

The concentration of glucose in your blood will start to rise after you have had a meal. Sugars from digested carbohydrate pass into the blood and are carried to the liver in the hepatic portal vein (Chapter 5). In the liver the glucose is converted to glycogen, so the blood leaving the liver in the hepatic vein has a lower concentration of glucose than when it enters the liver.

DIABETES

Some people have a disease where their pancreas cannot make enough insulin to keep their blood glucose level constant – it rises to very high concentrations. The disease is called **diabetes**. One symptom of diabetes can be detected by a chemical test on urine. Normally, people have no glucose at all in their urine. Someone suffering from diabetes may have such a high concentration of glucose in the blood that it is excreted in their urine. This can be shown up by using coloured test strips (Figure 7.5).



▲ Figure 7.5 Coloured test strips are used to detect glucose in urine.

KEY POINT

We should really refer to this disease by its full name, which is 'type 1' diabetes. There is also a 'type 2' diabetes, where the pancreas produces insulin but the body shows *insulin resistance*, where insulin has less effect than it should do. At first the pancreas makes extra insulin, but eventually it can't continue to make enough to maintain blood glucose at a normal level. Type 2 diabetes is common in people who are overweight and eat a poor diet that is high in sugar and other carbohydrates. It can be prevented and controlled by eating a good diet and doing regular exercise. Type 2 diabetes also tends to happen in middle-aged or older people, whereas type 1 can happen at any age, and is common in childhood.

Another symptom of diabetes is a constant thirst. This is because the high blood glucose concentration stimulates receptors in the hypothalamus of the brain. These 'thirst centres' are stimulated, so that by drinking, the person will dilute their blood.

Severe diabetes is very serious. If it is untreated, the sufferer loses weight and becomes weak and eventually falls into a coma and dies.

Carbohydrates in the diet, such as starch and sugars, are the source of glucose in the blood, so a person with diabetes can help to control their blood sugar if they limit the amount of carbohydrate that they eat. However a person

KEY POINT

Insulin is a protein, and if it were to be taken by mouth in tablet form, it would be broken down by protease enzymes in the gut. Instead it is injected into muscle tissue, where it is slowly absorbed into the bloodstream.

DID YOU KNOW?

Insulin for the treatment of diabetes has been available since 1921, and has kept millions of people alive. It was originally extracted from the pancreases of animals such as pigs and cows, and much insulin is still obtained in this way. However, since the 1970s, human insulin has been produced commercially, from genetically modified (GM) bacteria. The bacteria have their DNA 'engineered' to contain the gene for human insulin (see Chapter 22).

with diabetes (type 1) also needs to receive daily injections of insulin to keep the glucose in their blood at the right level.

People with diabetes can check their blood glucose using a special sensor. They prick their finger and place a drop of blood on a test strip. The strip is then put into the sensor, which gives them an accurate reading of how much glucose is in their blood (Figure 7.6). They can then tell when to inject insulin and how much to inject.



▲ Figure 7.6 Sensor for measuring blood glucose

CHAPTER QUESTIONS

More questions on chemical coordination can be found at the end of Unit 2 on page 130.

SKILLS CRITICAL THINKING 6

1 Which of the following statement(s) is/are true?

1. Insulin converts glucose to glycogen
2. Insulin causes blood glucose levels to fall
3. Glucose is stored as glucagon in the liver
4. Glycogen can be broken down to release glucose into the blood

A 2 only
B 2 and 4
C 3 and 4
D 1 and 2

BIOLOGY ONLY

2 Which hormone controls the development of the male secondary sexual characteristics?

A testosterone
B oestrogen
C progesterone
D follicle stimulating hormone (FSH)

END OF BIOLOGY ONLY

SKILLS CRITICAL THINKING 6

3 Which of the following will *not* happen when the hormone adrenaline is released?

A an increase in heart rate
B an increase in blood flow to the gut
C dilation of the pupils
D an increase in breathing rate

4 Which of the following is a symptom of type 1 diabetes?

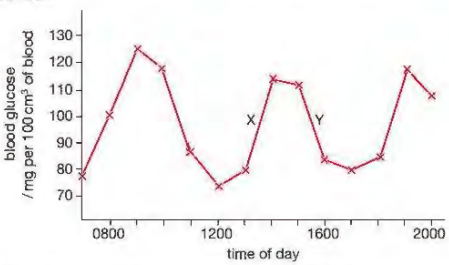
A high insulin level in the blood
B low glucose level in the blood
C glucose present in the urine
D insulin present in the urine

7 5 a *Hormones* are *secreted* by *endocrine glands*. Explain the meaning of the four words in italics

SKILLS ANALYSIS 6 b Identify the hormones A to D in the table.

Hormone	One function of this hormone
A	stimulates the liver to convert glucose to glycogen
B	controls the 'fight or flight' responses
C	controls the breaking of the voice at puberty in boys
D	completes the development of the uterus lining during the menstrual cycle

8 6 The graph shows the changes in blood glucose in a healthy woman over a 12-hour period.



a Explain why there was a rise in blood glucose at X.

b How does the body bring about a decrease in blood glucose at Y? Your answer should include the words insulin, liver and pancreas.

c Diabetes is a disease where the body cannot control the concentration of glucose in the blood.

SKILLS CRITICAL THINKING

i Why is this dangerous?

7 ii Describe two ways a person with diabetes can monitor their blood glucose level.

8 iii Explain two ways that a person with diabetes can help to control their blood glucose level.